

Research article

An innovative intrusion detection framework using GAN-augmented Deep Ensemble Neural Network for cross-domain IoT–cloud security

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ARTICLE INFO

Dataset link: <https://research.unsw.edu.au/projects/bot-iot-dataset>, <https://www.unb.ca/cic/datasets/ids-2018.html>

Keywords:

Machine learning/Deep learning
Internet-of-Things
Cloud computing
Vulnerabilities
Cyberattacks
Cross-domain
Cybersecurity
Multimodality

ABSTRACT

The rising popularity of smart cities and their impact across multiple sectors, including healthcare, transportation, and industry, is due to the rapid expansion of the Internet of Things (IoT). The growing popularity of IoT environments has made them susceptible to a wide array of cybersecurity hazards, such as denial-of-service (DoS), brute-force, and malicious access assaults. Robust intrusion detection and forensic investigation approaches must be developed to counter the aforementioned hazards. These frameworks primarily benefit from authentic and well-organized datasets for successful training and validation.

This study introduces an innovative Generative Adversarial Networks GAN-enhanced Deep Ensemble Neural Network (DENNW) framework designed specifically for cross-domain intrusion detection in cloud and IoT environments. This method significantly improves intrusion detection across various datasets by combining a multi-source learning architecture with GAN-based oversampling to address class imbalance. The Bot-IoT and CSE-CIC-IDS-2018 datasets are used in this research, containing both real and synthetic network traffic, covering a broad range of IoT and cloud-related incidents. The proposed GAN-based DENNW framework outperforms existing cloud-based approaches that use similar measures, providing comprehensive class-wise metric evaluation with 97.22% overall accuracy, surpassing many current studies. Although the DENNW framework achieves 93% accuracy with detailed class-wise analysis, the suggested approach enhances operational efficiency in the IoT sector. The results highlight that the proposed framework for protecting emerging IoT–cloud systems is adaptable and practical.

1. Introduction

The contemporary digital landscape, the growth of digital information has emerged as a critical component of modern society. However, the vast volume of data that travels across networks and systems is constantly exposed to cyberattacks and malicious

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<https://doi.org/10.1016/j.iot.2025.101773>

Received 16 June 2025; Received in revised form 15 September 2025; Accepted 22 September 2025

Available online 4 October 2025

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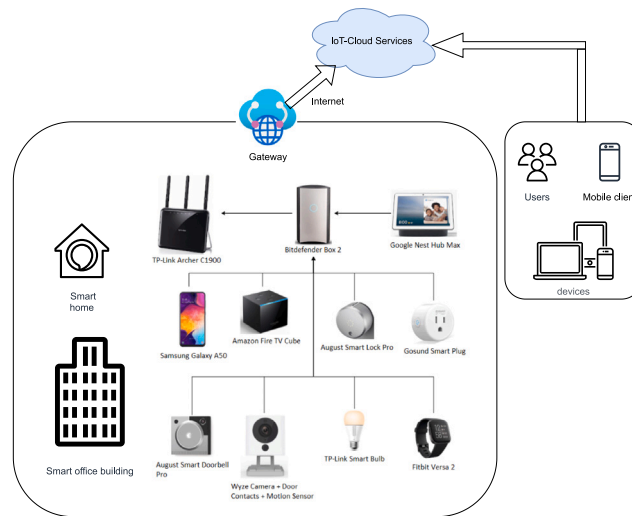


Fig. 1. Infrastructure of the IoT-Cloud Ecosystem.

endeavors that steal, compromise, or disrupt this invaluable data [1]. It is, therefore, imperative to enforce robust cybersecurity measures to safeguard against these threats [2,3].

The realm of cybersecurity is constantly evolving and encompasses a wide range of tools, techniques and protocols to protect our digital systems, networks, and data from potential harm. To uphold the reliability, adaptability, and authenticity of the digital world, it is imperative to remain vigilant against emerging cyberthreats [4,5]. The integration of an IoT-cloud based environment into various industries has revolutionized the generation, processing, and utilization of data. This era of interconnected technologies introduced new cybersecurity challenges like Phishing Attacks, Ransomware, Zero-Day Exploits, IoT-cloud Vulnerabilities [6–8]. As a result, businesses must adopt robust cybersecurity measures to protect their networks and data from potential threats [9,10].

As illustrated in Fig. 1, AI is a key component in IoT-cloud cybersecurity. However, AI-driven cybersecurity has limitations when handling individual, unimodal data types, such as sensor readings or structured cloud data. The challenge becomes even more significant in a cross-domain IoT-cloud environment due to the diverse data sources [11–13]. The current AI infrastructure faces difficulty providing security for the integrated environment, highlighting the urgent need for AI evolution. Advanced algorithms capable of interpreting various types of data cohesively are crucial for fortifying cybersecurity across interconnected IoT-cloud domains. Therefore, bridging this gap requires a transformative shift in AI capabilities [10,14].

Although GANs have been applied in various fields, their combination with deep ensemble models to address class imbalance in cross-domain IoT-cloud systems remains largely unexplored. This study presents a novel intrusion detection framework that effectively aligns heterogeneous IoT and cloud data into a unified DENNW, utilizing GAN-generated data to balance underrepresented attack classes. To the best of our knowledge, this is the first technique to improve detection robustness, accuracy, and generalizability by managing both GAN-augmented data imbalance and deep ensemble learning for cross-domain heterogeneity.

The remaining content of the article is organized into seven sections. Section 2 follows a research gap analysis, which is based on an investigation of relevant studies in cloud-based, IoT-based, and integrated IoT-cloud contexts. Experimental datasets are discussed in Section 3. The suggested methodology is explained in Section 4, which additionally addresses class-wise sample distribution, feature selection, data preparation, GAN-based oversampling for addressing class imbalance, and the DENNW training process. The experimental findings are presented and analyzed in Section 5. Afterwards, the shortcomings and challenges of the proposed framework are outlined in Section 6. This study ultimately concludes in Section 7, which also highlights prospective paths for future studies.

1.1. The necessity of a novel approach

The authors of [49,50] discuss the cybersecurity concerns for IoT applications in cloud computing domains. It is thus widely observed that as IoT progresses, cloud computing will play a pivotal role in receiving information from the various IoT devices. Furthermore, it is anticipated to serve as a foundation for data analytic, repositories, and universal data dispersion across multiple platforms. Prominent cloud providers like Microsoft Azure, Amazon Web Services (AWS), Google Cloud, etc., are proposed to launch communal access points for linking physical gadgets with processing and transmission abilities across multiple tools, techniques and amenities [51]. The existence of multiple access points introduces vulnerabilities that invaders can manipulate. These access points may exist within the communication perimeter of the network or be accessed distantly, either through wireless protocols or via the internet and cloud infrastructure. As a result, IoT-cloud networks are vulnerable to diverse malicious assaults [52–54]. Therefore, employing robust security measures to guard against such threats is indispensable. Due to resource limitations and software

Table 1

A systematic analysis of ML/DL techniques and datasets against a variety of cyberattacks in the cloud-based context.

Ref	Dataset	Technique	Research gap		Accuracy	Attack
			Unimodal	Multimodal		
[15], 2020	Self-created dataset	DT, RF, and LR	✓	□	97.9%	Malware attacks
[16], 2024	SOREL-20M and EMBER-2018	CNN framework	✓	□	96.9%	
[17], 2019	UNSW, ISOT, TOR	CNN	✓	□	98.6%	Data traffic analysis
[18], 2022	FRAUD1, FRAUD2	Hybrid ConvLSTM	✓	□	99%	Spoofing attacks
[19], 2016	DARPA98 and KDD-Cup99	Cloud-based ML technique	✓	□	good	Anonymous web proxies
[20], 2020	Create own dataset	Ridge Regression	✓	□	good	Coresident attack
[17], 2019	DRAPA, KDD, UNSW and ISOT	CNN	✓	□	good	Intrusion attacks
[21], 2024	UNSW-NB15	PCA, CNN	✓	□	100%	
[22], 2021	KDDCup99, Kyoto 2006+, NSL-KDD, UNSW-NB15	ML and DL frameworks	✓	□	good	
[23], 2021	Real time datasets, IDS datasets	ML, Hybrid Meta-Heuristic technique	✓	□	63.64%, 80.00%	
[24], 2024	Own dataset	KNN, RF, DT and ML techniques	✓	□	99.59%	
[25], 2024	CICIDS 2017	DNN	✓	□	98.39%	Distributed Denial of Service
[26], 2021	Genuine traffic datasets, TON-IoT and CIC-DDoS2019	CNN, DNN, RNN	✓	□	95.12%	

compatibility issues, IoT–cloud infrastructures stipulate diverse security integration challenges compared to conventional local area networks. The complexity is augmented by the dynamic landscape of the IoT–cloud system, which enables modifications to be made to data profiles, device composition, settings, and user behavior patterns [55]. Hence, AI-based cross-domain security solutions are becoming increasingly important due to the rapid expansion and widespread acceptance of IoT devices and the interlinked nature of IoT–cloud infrastructure, which complicates IoT–cloud security challenges. Despite the efficacy of prevailing techniques in detecting cyberattacks in isolated environments, it remains challenging to identify these assaults in a cross-domain integrated infrastructure. To tackle these security challenges, a novel methodology that considers the complexities of IoT–cloud amalgamation is necessary. In addition to diverting potential attacks, this technique must rapidly and accurately address the detection and elimination of threats. This capability is imperative for ensuring the security of end devices from the perspectives of both individual end-users and operators/providers managing the entire IoT ecosystem [6–8].

1.2. Scope of study

The interdependence and vulnerability of IoT and cloud environments have become crucial areas of research for AI applications in cybersecurity. Cloud systems serve as central hubs for storing and analyzing vast amounts of data, while IoT devices generate diverse data and are exposed to various threats, leading to increased cybersecurity risks [56]. A key focus has been on developing adaptive strategies using ML, DL and AI to tackle the complex and ever-changing cyberthreats in these domains. The aim is to create innovative cybersecurity solutions that can counter the range of sophisticated attacks prevalent in the present digital ecosystem, safeguarding cloud and IoT systems against potential threats [50,57].

This research consists of five sections. The study begins with an introduction that provides an overview of its purpose, motivation, necessity, scope, and contribution to the research. Subsequently, the related work section provides a detailed analysis of existing research contributions to AI-based methodologies used to counter numerous attacks against IoT and cloud domains. Furthermore, the dataset exploration section provides a complete investigation of the datasets used in this research. Later, a detailed explanation in terms of data-preprocessing steps, oversampling techniques and experimental details is discussed in the experimental analysis section. Finally, the article concludes with a discussion of the results and conclusion sections.

Table 2

A systematic analysis of ML/DL techniques and datasets against a variety of cyberattacks in the IoT-based context.

Ref	Dataset	Technique	Research gap		Accuracy	Attack
			Unimodal	Multi-modal		
[27], 2019	Maling VirusSign and VirusShare	ML, DL frameworks	✓	□	94.0, 94.8%	Zero-day malware
[28], 2021	DARPA, KDD Cup 99, NSL-KDD, UNSW-NB15, DREBIN	DL and NN methods	✓	□	high	malware, botnet and phishing
[29], 2019	CICIDS2017	DL models	✓	□	97.16%	DDoS
[30], 2024	MQTT-IoT-IDS2020	KNN	✓	□	95%	DDoS
[31], 2018	DARPA KDDCUP99	DL schemes	✓	□	99.20%	Probing, R2L, U2R
[32], 2020	UNSW-NB15, CICIDS2017	LR, SVM, DT, RF	□	✓	high	cyberattacks
[33], 2020	IoT-23, LITNET-2020, and NetML-2020	DNN and LSTM	✓	□	96.93%	Web/network attacks
[34], 2022	CIDCC-15, UNSW-NB15, and NSL-KDD	LSTM	✓	□	98.89%	
[35], 2020	CSIC 2010, FWAF, HttpParams	DL frameworks	✓	□	99.410%	
[36], 2015	Bot-IoT dataset	SMOTE-DRNN	✓	□	good	Botnet detection
[37], 2021	UNSW-NB15, CICDDoS2019, ISCX 2012	ML and DL algorithms	✓	□	99.9%	Intrusion attacks
[38], 2022	NSL-KDD dataset	Hybrid DL model	✓	□	94.69%	
[39], 2021	Self-created dataset	FFNW	✓	□	99.79%	
[40], 2024	NSLKDD	FFNW	✓	□	96.36%	
[41], 2019	CIC-Darknet2020	Hybrid model	✓	□	high	Darknet intrusion detection
[42], 2020	CXVPN2016, ISCXTor2017	Hybrid DL model	✓	□	86%	

2. Related work

AI and ML have been developed to enable machines to learn and think like humans. As a result, AI has made revolutionary achievements in various fields such as object detection, image classification, speech identification, Natural Language Processing (NLP), cybersecurity, and many others [58–61]. Progress in ML, which is a fundamental component of AI, has become an essential tool to analyze user information and improve cybersecurity technologies and applications [14]. Cybersecurity concerns are the driving force behind many of the advances in AI and ML [62]. Additionally, data science has become a leading innovative scientific representation, incorporating a variety of techniques such as textual analysis, spam filtering, video assessment, picture identification, and others [28,63,64]. DL, one of the essential ML algorithms, has been shown to outperform its predecessors and is suitable for cybersecurity-related applications [4,65]. Fig. 2 categorizes AI-based solutions into probing, User to Root (U2R), Root to Local (R2L), and Denial-of-service (DoS) approaches for cybersecurity concerns. The probe assaults are designed to acquire data from sources that vary from region to region, such as port sweeps and IP sweeps. Probe attacks make information in peer networks vulnerable to intruders, letting them snoop, access, or acquire data. This hazard may be detected using AI-based technologies [66]. The U2R assault aims to get access to resources as if they were normal users. U2R attacks can be used to influence, spy on, or interrupt the normal operation of a computer system. A unique SVM model based on a security architecture was suggested [67–69] to handle these attacks. In the case of an R2L attack, a client transmits messages to a machine to which they do not even have complete rights. R2L attacks make use of default credentials. In [70,71], the scholars suggested two AI-based approaches for identifying R2L

Table 3

A systematic analysis of ML/DL techniques and datasets against a variety of cyberattacks in the integrated IoT-cloud based context.

Ref	Dataset	Technique	Research gap		Accuracy	Attack
			Unimodal	Multimodal		
[43], 2022	DARPA	AdaBoost, RF, DT, SVM, MultinomialNB	✓	□	99.8%	Network attacks
[44], 2023	NSL-KDD	KNN, RF, DT, SVM, Ensemble	□	✓	good	
[45], 2023	Heart diseases dataset	C-Means clustering along with customized Bi-LSTM	□	✓	89.67%	DDoS and zero-day
[46], 2020	NSL-KDD	Proposed Sparse AE	✓	□	90%	DoS, R2L, U2R, Probe
[47], 2023	EMNIST, X-IIoTID, and Federated TON-IoT	DT, RF, SVM, DNN and Transfer Learning (CMTL) algorithm	✓	✓	high	DoS/DDoS, Malware, Injection and Man in the Middle attacks
[48], 2023	UNSW-NB15 and NSL-KDD	Ensemble Bagged Trees Detection (EBTD) algorithm	✓	□	high	malicious behaviors

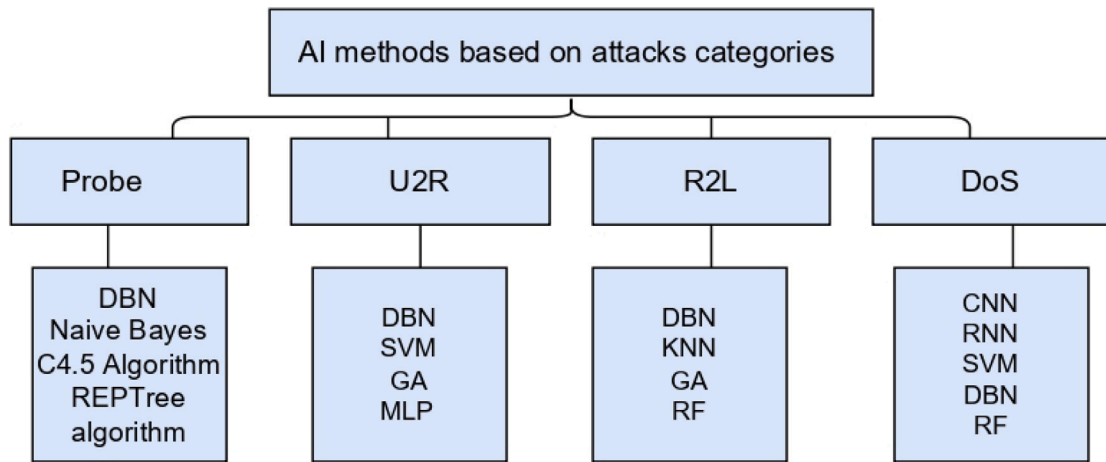


Fig. 2. AI-based methods categorization against various types of cyberattacks.

assaults. Another use of AI is to detect a DoS attack. Due to their simplicity of deployment, denial-of-service (DoS) assaults are highly popular [72]. DDoS and UDP storms, for example, can be used to interrupt data transit. DoS attacks overload the network capacity, making it harder to handle legitimate networking requests. An AI recognition system based on numerous ML/DL algorithms such as Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and SVM has been proposed to identify DoS attacks for the IoT Botnet dataset [73–76].

This section comprises three subheadings that comprehensively explore the use of classic ML and DL approaches for cybersecurity in different environments, such as IoT and cloud-based scenarios, individually, as well as in an integrated IoT-cloud based environment [77–79]. Furthermore, the investigations conducted by other researchers are summarized in Table 1, Tables 2 and 3 based on datasets, cyberattacks, and ML and DL techniques.

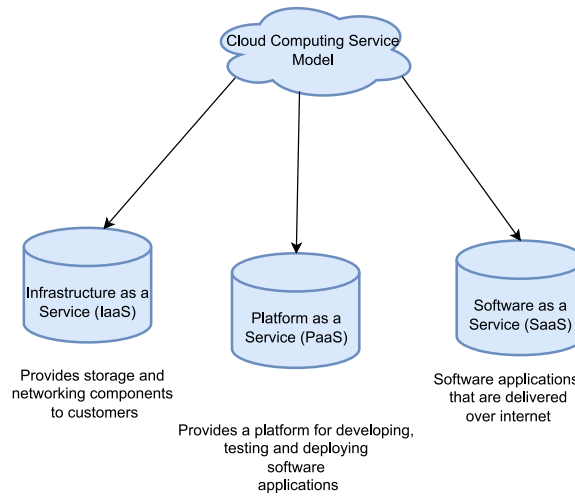


Fig. 3. Infrastructure as a Service (IaaS) simplifies the digitalized computing framework, while Platform as a Service (PaaS) provides a dedicated platform for development. Additionally, Software as a Service (SaaS) renders software applications over the network, permitting end users to access them without the dependence on local installation.

2.1. Related work for cloud-based context

The proliferation of the World Wide Web and the advancement of sensors, computing devices, and electronic gadgets have led to a significant transformation in the technological deployment of consumers around the world. However, most real-time applications face significant challenges, such as achieving high performance with minimal delay and reducing costs associated with service utilization due to the growing requirements of memory and computing resources of smart gadgets. In such a scenario, cloud computing can be a valuable tool to provide customers with streaming services that are both timely and cost-effective [80].

Cloud computing (CC) provides on-demand access to a pool of configurable IT resources, including servers, applications, data storage, and network connectivity. This enables businesses to swiftly deploy these resources with minimal administrative effort and service provider involvement. Despite its many benefits, security remains a significant challenge for organizations that adopt cloud-based technology [26]. Cloud technology is inherently decentralized and publicly accessible, which makes it more vulnerable to attacks and risks. Attackers are emboldened by the ease with which they can target systems that rely on cloud-based data. Such assaults can compromise the reliability, security, and accessibility of cloud infrastructure and assets. Cloud architecture is essentially a platform that allows for the delivery of services through the internet. It offers a remarkable degree of elasticity, allowing organizations to modify their resource requirements in response to changing needs, which can significantly reduce costs. The services can be delivered in the form of software or shared infrastructure over the web [81,82]. The CC service model showcased in Fig. 3 provides users with three types of services: platform as a service (PaaS), infrastructure as a service (IaaS), and software as a service (SaaS). Users can submit their private data and corporate policies to all these services. To maintain customer loyalty, it is critical to secure private data and corporate regulations. Therefore, security measures must be implemented in SaaS, PaaS, and IaaS [83–85]. These security measures must ensure minimal risk from unauthorized access, authorization for access, links or routes, information, and information preservation in addition to the services provided [17,86]. SaaS allows customers to access and use applications and services provided by a vendor. PaaS refers to the ability of a vendor to develop and deploy software using their programming languages, libraries, services, and tools. The vendor manages and controls the underlying infrastructure while the customer controls the installed programs and the parameters of the hosting environment [80,87]. IaaS allows users to allocate computing resources to install and run any application, including operating systems. Customers have some control over the platform and memory to be utilized [81,88]. The need for cloud security has become paramount in the modern world, where sensitive data is uploaded to the cloud. AI-based techniques are effective in safeguarding the cloud environment against cyberattacks and privacy violations [17,89,90].

Table 1 depicts a comparative analysis of existing literature on implementing ML/DL techniques for cyberattack detection, concentrating on the datasets utilized. This study identifies research gaps by focusing on unimodal and multimodal scenarios. Based on the reviewed literature, it is clear that most authors utilize diverse unimodal AI-based methods for cyberattack identification in the cloud-based scenario. Especially, the authors of [91,92] have studied limited multimodality, predominantly by detecting multiple types of attacks for cloud-based environments instead of employing varied data types for threat identification.

2.2. Related work for IoT-based context

The Internet of Things (IoT) is a network of interconnected physical devices, algorithms, and sensors that communicate with each other over the Internet [2,93]. These devices are equipped with various technologies such as electrical chips and RFID tags, allowing

for unique identification and remote monitoring and control. The IoT offers ubiquitous connectivity to smart objects, cloud-based services, and other instances, making it a vast ecosystem with millions of diverse devices and stakeholders. The intelligent nature of these devices enables them to interact with other connected components and provide real-time data, making them an integral part of modern technology [94,95].

IoT technology is increasingly being adopted to create smart communities, educational institutions, e-commerce platforms, and digital banking systems, among others. However, as IoT devices are constantly connected to the internet, they are susceptible to open attacks, which can compromise their reliability, security, and accessibility. Regardless of their public nature, IoT gadgets are vulnerable to various hazards that can threaten their integrity. With the growth of IoT, the frequency and sophistication of attacks are also increasing. As a result, protecting these devices in such a diverse context has become a challenging task, requiring more advanced security measures [96]. Attack detection strategies can be broadly classified into two types: signature-based and anomaly-based. In the signature-based approach, the incoming traffic is compared against a repository of known attack types. On the other hand, the anomaly-based approach identifies attacks by detecting deviations from the typical traffic patterns [92]. While the signature-based approach has shown remarkable detection accuracy and relatively low false alarms, it has been criticized for its inability to detect innovative attacks. Conversely, anomaly detection may not have sufficient precision, but it is capable of identifying novel attacks. Both approaches have been widely used in attack detection with the application of conventional ML techniques [12,38,97].

The use of AI-based tools for identifying vulnerabilities has become more common in recent years. ML and DL are recognized as valuable technologies for preventing and managing cyberthreats. ML algorithms are particularly suited for IoT security due to their adaptability, reliability, and configurability. They can generate diagnoses on new input without requiring individuals to hard-code trends. Instead, they learn from existing information to identify trends and produce diagnoses. However, even conventional ML models struggle to handle the complexities of modern IoT infrastructures and emerging network vulnerabilities. These models often reach an accuracy limit, need hand-crafted features, and cannot handle large datasets [98–100]. Sophisticated cyberattacks are becoming increasingly difficult to detect using traditional ML techniques. This is due to the constantly evolving resources and capabilities of attackers. About 99% of cyberattack variants are minor variations of known hacking attempts. Even the remaining 1% of unique threats are based on previous theories and logical premises. Therefore, conventional ML methods are unable to identify subtle characteristics that differentiate new attacks or variants from normal ones. This limitation makes it challenging to detect minor changes that could indicate an attack [28,101].

DL models have become increasingly popular in cybersecurity due to their ability to handle large amounts of data, identify features on their own and perform better than traditional ML models in terms of accuracy and efficiency. DL models are reliable in identifying the true nature of digital information, even with slight modifications, and can produce exceptional permanent representations of the learning data. Recent research has indicated the potential of DL models in detecting cybersecurity attacks, particularly in the context of the IoT, which generates massive amounts of data dealing with private information [102,103].

The authors of [104] presented an integrated federated learning system optimized for industrial IoT malware detection, with a focus on decentralized model training. Table 2 depicts a systematic analysis of ML/DL-based techniques for identifying attacks in IoT systems. This exploration exposes the lack of multimodal framework implementation for cyberattack detection in IoT infrastructure and highlights the research work of other scholars concentrating on unimodal cyberattack detection.

2.3. Related work for integrated IoT-cloud based context

IoT and Cloud environments are becoming increasingly interwoven in the contemporary technological landscape [55]. These integrated domains, which comprise multiple systems of devices distributed over distant localities, operate within an environment that is not always absolutely reliable. There are exceptional challenges and security hazards associated with this setup that must be thoroughly dealt with. One of the key strengths of IoT amalgamation with a cloud environment is the requirement for efficient information processing and seamless association across numerous devices [105]. For example, a Smart Home system usually consists of multiple microcontrollers furnished with actuators and sensors to assemble, process, and operate various physical elements of infrastructure. These devices bind locally and depend on cloud services for both core and auxiliary functions like data processing, storage, and remote access [43].

Table 3 demonstrates the comprehensive analysis of existing research on the integrated cross-domain multimodal IoT-cloud environment. It indicates that although some scholars, such as those referenced in [43,46], and [48] have explored interdomain systems, their investigations lacked complex analysis of different data types. Furthermore, some researchers have investigated multimodal data within isolated scenarios. Significantly, the authors of [106] focused on both cross-domain systems but with a limited scope of multimodality. These findings support the research gap that this study strives to address.

2.4. Research gap analysis

The integration of DL and ML has revolutionized the field of technological advancements, providing breakthrough solutions across various industries. An exhaustive investigation of AI-based cyberattack detection methods for isolated cloud and IoT environments is demonstrated in Tables 1 and 2. However, despite the abundance of research on DL and ML applications, there still exists a significant research gap when it comes to effectively deploying these technologies in multi-environment scenarios [107]. Although, there are several successful deployments of these technologies in individual environments, there is a lack of systematic research on their use in cross-domain settings. This study aims to identify and address this gap by conducting a comprehensive analysis, which leads to the development of a viable solution [54,108]. Hence, the research topic of the current study centers on the development and implementation of a cross-modal DL framework within an integrated IoT-cloud infrastructure.

Table 4
Traffic type distribution in BoT-IoT dataset.

Category	Type	Samples
Information Gathering	Service Scanning	1463,364
	OS Fingerprinting	358,275
Denial of Service (DDoS)	TCP	19,547,603
	UDP	18,965,106
	HTTP	19,771
Denial of Service (DoS)	TCP	12,315,997
	UDP	20,659,491
	HTTP	29,706
Information Theft	Keylogging	1,469
	Data Theft	118
Total		73,360,900

3. Dataset exploration

This study introduces an intrusion detection framework for IoT and cloud environments; hence, separate datasets for respective domains are used for experimental analysis. Bot-IoT and CSE-CIC-IDS-2018 datasets.

BoT-IoT dataset: It has been generated by establishing an authentic networked system in the UNSW Canberra Cyber Center's Cyber Range Lab. A mixture of standard and botnet traffic has been detected in this environment. The raw files for the dataset are available in a multitude of ways, such as the resulting Argus files, the preliminary PCAP files, and CSV files. The data were divided into attack categories and subcategories to simplify the identification technique. There are over 72,000,000 instances in the 69.3 GB collected PCAP files. The derived flow traffic is 16.7 GB in size in a CSV file format. DDoS, DoS, OS and Service Scan, Keylogging, and Data Exfiltration threats are all featured in the dataset. The DDoS and DoS attacks have been further organized according to the protocol used. This traffic type distribution is depicted in Table 4.

CSE-CIC-IDS-2018 dataset: A combined partnership between the Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC) generated the data available for the CSE-CIC-IDS-2018. It was designed for examining research investigations on intrusion detection systems and is now a standard dataset for assessing IDSs. A broad set of possibilities for investigation can be obtained from this dataset, which has been rigorously examined and made to mimic actual digital threats and intrusions. The potential to simulate intricate system settings proves valuable since it permits researchers and professionals to examine and strengthen security measures.

Table 5 illustrates the statistics obtained over ten days, having eighty features. There are fifteen categories of attacks: FTP-Brute Force, SSH-Brute Force, DoS attacks-GoldenEye, DoS attacks-Slowloris, DoS attacks-Hulk, DoS attacks-SlowHTTPTest, DDoS attacks-LOIC-HTTP, DDOS attack-HOIC, DDOS attack-LOIC-UDP, Brute Force-Web, Brute Force-XSS, SQL injection, infiltration, label, and bot.

4. Research methodology

This section provides complete methodological details of the research performed by the efficient and intelligent GAN-Augmented DL-based intrusion detection framework for cross-domain scenarios. The analysis illustrated in Fig. 4 depicts the operational workflow of proposed framework. The GAN component is effectively integrated into the IoT data analysis pipeline in the proposed structure (Fig. 4) to address the significant class imbalance before training through the DENNW. Since IoT traffic is domain-dependent, the GAN is trained from scratch to generate artificial samples for minority classes using the top 15 features selected by Neighborhood Component Analysis (NCA) from the IoT dataset. The Adam optimizer is used to systematically train the generator and discriminator until the discriminator judges that the generator produces high-quality samples indistinguishable from real data. The DENNW architecture is then adjusted with specific hyperparameters and trained from scratch on a balanced, preprocessed IoT and Cloud dataset to maximize the use of the best features. By optimizing detection efficiency across all attack types, this seamless integration of ensemble learning and GAN-based oversampling ensures that the final model is trained on a balanced, feature-rich, and multi-source dataset.

Fig. 5 demonstrates the structural pattern of a sophisticated and innovative DENNW model that has been designed to deal with demanding computational jobs. This experimental approach reveals that it can navigate data across different layers, all assist with feature extraction, transformation, and classification. The structure commences with a feature input layer that handles fifteen/full attributes for respective datasets and travels across several fully connected layers that have been detached by batch normalization and ReLU activation functions. This guarantees that features are captured efficiently and addresses challenges like vanishing gradients. Additionally, its capacity to determine temporal dependencies is emphasized by the incorporation of an LSTM layer with 128 hidden units, which enables it for sequential processing of the information. Subsequently, the cross-entropy loss estimation reveals that the SoftMax layer and classification output layer allow efficient categorization into different classes. This systematic methodology demonstrates the efficacy and versatility of the framework in managing a variety of datasets while retaining its excellent precision and adaptability.

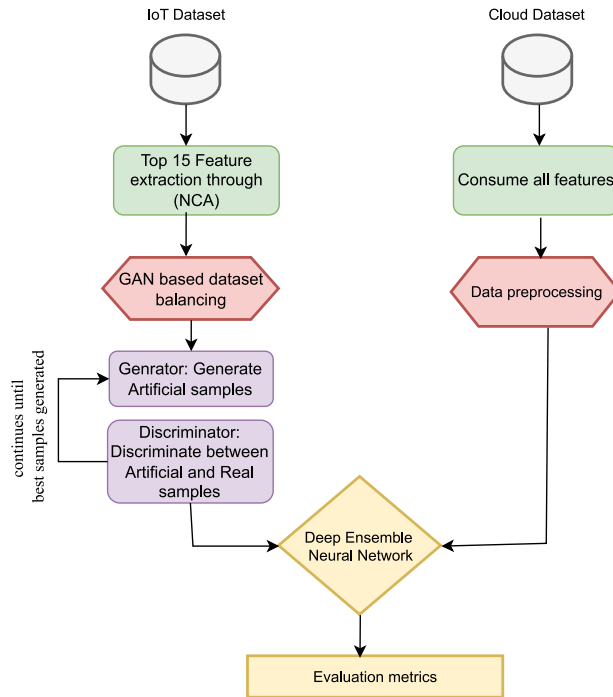


Fig. 4. System flow diagram of proposed GAN-based DENNW for IoT-cloud cross-domain environment.

ANALYSIS RESULT				
	Name	Type	Activations	Learnable Prope...
1	featureinput 15 features	Feature Input	15(C) × 1(B)	-
2	fc_1 256 fully connected layer	Fully Connected	256(C) × 1(B)	Weights 256 × 15 Bias 256 × 1
3	batchnorm-2_1 Batch normalization	Batch Normalization	256(C) × 1(B)	Offset 256 × 1 Scale 256 × 1
4	relu-2_1 ReLU	ReLU	256(C) × 1(B)	-
5	fc_2 256 fully connected layer	Fully Connected	256(C) × 1(B)	Weights 256 × 15 Bias 256 × 1
6	batchnorm-3 Batch normalization	Batch Normalization	256(C) × 1(B)	Offset 256 × 1 Scale 256 × 1
7	relu-3 ReLU	ReLU	256(C) × 1(B)	-
8	fc_3 256 fully connected layer	Fully Connected	256(C) × 1(B)	Weights 256 × 15 Bias 256 × 1
9	batchnorm-1 Batch normalization	Batch Normalization	256(C) × 1(B)	Offset 256 × 1 Scale 256 × 1
10	relu-1 ReLU	ReLU	256(C) × 1(B)	-
11	addition Element-wise addition of 3 inputs	Addition	256(C) × 1(B)	-
12	fc_4 128 fully connected layer	Fully Connected	128(C) × 1(B)	Weights 128 × 256 Bias 128 × 1
13	batchnorm-4 Batch normalization	Batch Normalization	128(C) × 1(B)	Offset 128 × 1 Scale 128 × 1
14	relu-4 ReLU	ReLU	128(C) × 1(B)	-
15	fc_5 64 fully connected layer	Fully Connected	64(C) × 1(B)	Weights 64 × 128 Bias 64 × 1
16	batchnorm-5 Batch normalization	Batch Normalization	64(C) × 1(B)	Offset 64 × 1 Scale 64 × 1
17	relu-5 ReLU	ReLU	64(C) × 1(B)	-
18	fc_6 32 fully connected layer	Fully Connected	32(C) × 1(B)	Weights 32 × 64 Bias 32 × 1
19	batchnorm-6 Batch normalization	Batch Normalization	32(C) × 1(B)	Offset 32 × 1 Scale 32 × 1
20	relu-6 ReLU	ReLU	32(C) × 1(B)	-
21	flatten Flatten	Flatten	32(C) × 1(B)	-
22	lstm LSTM with 128 hidden units	LSTM	128(C) × 1(B)	InputWeight... 512 × ∞ RecurrentWe... 512 × ∞ Bias 512 × ∞
23	fc 5 fully connected layer	Fully Connected	5(C) × 1(B)	Weights 5 × 128 Bias 5 × 1
24	softmax softmax	Softmax	5(C) × 1(B)	-
25	classoutput crossentropyloss	Classification Output	5(C) × 1(B)	-

Fig. 5. Structural analysis of Deep Ensemble Neural Network [109].

Table 5
Traffic type distribution in CSE-CIC-IDS-2018 dataset.

Broad category	Traffic type (Class Labels)	Number of samples	Category total
Benign	Benign	13,484,708	13,484,708
	–	–	
	Normal (Benign)	5,329,008	
	–	–	
DDoS	DDoS attacks- LOIC-HTTP	575,364	775,955
	DDoS attack-HOIC	198,861	
	DDoS attack-LOIC-UDP	1,730	
	–	–	
DoS	DoS attacks-Hulk	145,199	196,568
	DoS attacks- GoldenEye	41,406	
	DoS attacks-Slowloris	9,908	
	DoS attacks- SlowHTTPTest	55	
BruteForce	SSH-Bruteforce	94,048	94,898
	Brute force -Web	568	
	Brute force -XSS	229	
	FTP-BruteForce	53	
Bot	Bot	144,535	144,535
Infiltration	Infiltration	118,483	118,483
SQL Injection	SQL Injection	85	85
Anomaly (All Attacks)	(Sum of All Attack Types)	1,330,524	6,659,532
Total	(Benign + Anomalies)	6,659,532	

The experiment is conducted on the Windows 11 platform using MATLAB 2022a as a development environment. A Dell Core i5 7th generation machine with 16 GB RAM and 512 GB SSD is used for experimental purposes. This section comprises the following steps:

4.1. Data preprocessing

The bedrock for the research investigation had been created by the data preprocessing treatment technique, aiming to make sure that the raw data is structured, organized, and suitable for further manipulation. As both of the datasets are too huge and to accommodate the lack of resources 10% of the datasets are selected. The existence of the significant column “category”, which refers to the tags used for labeling, can be verified by carefully examining the dataset. Afterward, the data is stripped of extraneous columns, particularly the ones that are blank. The “category” column and just the numerical columns are retained. A systematic sample strategy was employed, adopting 10% of the raw data for investigation to eliminate potential imbalances in the dataset. To eliminate over-representation for the Bot-IoT dataset, the sample selection approach examined larger classes like DoS and DDoS proportionately while ensuring that less prevalent classes like Theft, Normal, and Reconnaissance are sufficiently captured.

4.2. Feature selection using neighborhood component analysis (NCA)

A supervised dimensionality reduction method called NCA has been employed for Bot-IoT dataset while selecting the best features from the dataset to optimize it for training. NCA is particularly beneficial for lowering noise and inconsistency by determining characteristics that are especially relevant to the prediction process. This phase strengthened the usefulness of the dataset and boosted the impact of further predictive models. The SGD (Stochastic Gradient Descent) solver has been utilized to train the NCA model, guaranteeing optimal efficiency. To optimize efficacy and technological advancement effectiveness, variables encompassing the number of tuning iterations, pass limit, and subset size were carefully adjusted and illustrated in Table 8. To find the most significant traits such as the NCA-generated feature weights have been organized in descending order following training. The best 15 traits are chosen from these, substantially decreasing the computational complexity of the dataset while conserving pertinent data. This phase boosted the generalization of the model by dropping the likelihood of overfitting and guaranteeing that only the most appropriate traits are utilized for training, in addition to enhancing the dataset.

Table 6

Original and target class distributions after applying GAN-based over-sampling for BoT-IoT dataset.

Class	Original samples	GAN generated samples
Normal	973	500 000
Theft	139	200 000
Reconnaissance	182 296	300 000
DoS	3300 519	–
DDoS	3853 117	–

Table 7

Sample distribution per class for CSE-CIC-IDS2018 Dataset.

Class	Number of samples
Benign	6205,356
DDoS attack-HOIC	198,861
DDoS attacks-LOIC-HTTP	575,364
DoS attacks-GoldenEye	41,406
DoS attacks-Hulk	145,199
DoS attacks-Slowloris	9,908
SSH-Bruteforce	94,048

Table 8

Comparison of parameters across different techniques.

Parameter	NCA parameters	GAN parameters	DNNW parameters
Solver	sgd	–	–
Number of Features	15	–	–
Tuning Subset Size	100	–	–
Number of Tuning Iterations	5	–	–
Pass Limit	5	–	–
Generator Hidden Size	–	128 neurons	–
Activation Functions	–	ReLU, Tanh	–
Batch Size	–	32	150
Epochs	–	20	10
Optimizer	–	Adam	Adam
Learning Rate	–	0.0002	0.001
Validation Frequency	–	–	20 iterations
Data Split Ratios	–	–	Train (70%), Validation (15%), Test (15%)

4.3. GAN-based oversampling for handling class imbalance

The effectiveness of the study relies on managing class imbalance, a prevalent issue in ML and DL ventures. The imbalanced nature of the dataset may lead to biased models and skew the results towards the majority classes. The BoT-IoT dataset has been balanced by using GAN-based oversampling to produce artificial instances for poorly represented categories like Theft, Normal, and Reconnaissance. This technique prevents the framework from becoming biased towards DoS and DDoS, the two most prevalent classes. The initial stage in this technique involves figuring out the optimal equilibrium and desirable quantity of instances for each class.

The GAN architecture comprises two main sources: a generator and a discriminator. The Generator tends to generate artificial specimens that are analogous to the patterns of real data, and the Discriminator subsequently strives to distinguish between fabricated and real data. The Generator has the potential to generate trustworthy information more effectively over time due to this training technique. The optimal Eq. (1) is employed to illustrate the computational structure of a GAN:

$$\min_G \max_D \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))] \quad (1)$$

Where:

- $G(z)$: The random noise z is generated in response to the generator output experimented from the preceding distribution $p_z(z)$ (usually Gaussian or uniform)
- $D(x)$: The Discriminator calculates the probability of x drives from the actual dissemination of data $p_{\text{data}}(x)$.
- $p_{\text{data}}(x)$: The actual dissemination of data.
- $p_z(z)$: The noise dissemination is provided as input to the Generator.

The training process encompasses rotating updates to the Generator and Discriminator. The Discriminator exploits its capacity to discriminate between actual instances x from artificial instances $G(z)$. The Generator reduces the competence of the Discriminator to discriminate artificial instances from actual ones.

Generator Network: The Generator accepts random noise z (derived from a uniform distribution $z \sim \mathcal{U}(-1,1)$) as input. It is comprised of an output layer with a tanh activation function for normalization of the output, two hidden layers with 128 neurons each leveraging ReLU activation functions, and an input layer that is aligned with the feature size. The Adam optimizer has been employed to train the GAN for 30 epochs with a batch size of 32, guaranteeing that the samples obtained are closely correlated with the actual data. The GAN delivered an adequate number of artificial instances for each poorly represented class, which later got added to the original dataset to yield an ultimate dataset that is well-balanced. The balance attained through this procedure was confirmed by a bar graph that displayed the class distribution after oversampling.

Discriminator Network: It is a binary classifier that grabs actual or artificial instances as input. It consumes fully connected layers with Leaky ReLU activations and a sigmoid output layer to forecast the probability that a provided instance is actual. The loss functions for the Generator and Discriminator are depicted in Eq. (2) and Eq. (3):

Generator loss:

$$L_G = -\mathbb{E}_{z \sim p_z(z)} [\log(D(G(z)))] \quad (2)$$

Discriminator loss:

$$L_D = -\mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] - \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))] \quad (3)$$

GAN Training Steps The artificial specimens are subsequently developed by following the steps of the generator.

1. Generator Update:

- Produce artificial instances $G(z)$.
- Calculate $D(G(z))$ for these instances.
- The weights of Generator are updated to optimize L_G , efficiently “fooling” the Discriminator.

2. Discriminator Update:

- Calculate $D(x)$ for actual instances x from the data.
- Calculate $D(G(z))$ for artificial instances $G(z)$ produced by the Generator.
- The weights of the Discriminator are updated to optimize L_D .

4.4. Class-wise sample distribution

This research investigates the potential to enhance data augmentation for Generative Adversarial Networks (GANs), which encompass various types, including Conditional GANs (cGANs), Wasserstein GANs (WGANs), and Deep Convolutional GANs (DCGANs). Instead of conducting a comprehensive comparison of different GAN structures, the main goal of this study is to demonstrate that using GAN-based augmentation to address class imbalance in minority attack classes is feasible and advantageous. Therefore, the Vanilla GAN, the simplest form of GAN, is used to verify that synthetic oversampling can enhance the overall effectiveness of the proposed intrusion detection system. Hence, this oversampling approach has been successfully applied to address class imbalance in the BoT-IoT dataset. GAN is trained for 30 epochs with a batch size of 32 and an Adam optimizer (*learning rate* = 0.0002, $\beta_1 = 0.5$), subsequently, it fabricated artificial instances for each under-represented class. These generated instances are concatenated with the original dataset to produce the final balanced dataset. The Table 6 provides details on the actual and GAN-generated distribution of samples for the BoT-IoT dataset. The adversarial training ensured that the synthetic samples were indistinguishable from real data, enhancing the performance of the model for under-represented classes. The class-wise sample distribution after GAN-based oversampling is depicted in Table 6 and Fig. 6 for the BoT-IoT dataset.

The following classes (Benign, DDOS attack-HOIC, DDOS attacks-LOIC-HTTP, DoS attacks-GoldenEye, DoS attacks-Hulk, DoS attacks-Slowloris and SSH-Bruteforce) are selected from the CSE-CIC-IDS2018 dataset after investigating the whole data distribution. These exclusive classes are selected to ensure sufficient data for effective model training and evaluation. Each designated class preserves its complete set of attributes to maintain the integrity and contextual relevance of the data. Table 7 and Fig. 7 depict the tabular and visual representation of class-wise sample distribution for the CSE-CIC-IDS2018 dataset that highlights their proportion in the dataset against selected classes.

4.5. Training the Deep Ensemble Neural Network (DENNW)

The DENNW has been trained using the cleaned and balanced dataset. This stage aims to create an effective framework that accurately identifies multiple categories of cyberthreats. Initially, 10% of the dataset is categorized into 70-15-15 percent for training, validation and testing. The 10% portion of the GAN-generated dataset was created using simple random selection, confirming that each record had an equal chance of being selected without explicitly applying class label stratification. A Chi-square goodness-of-fit

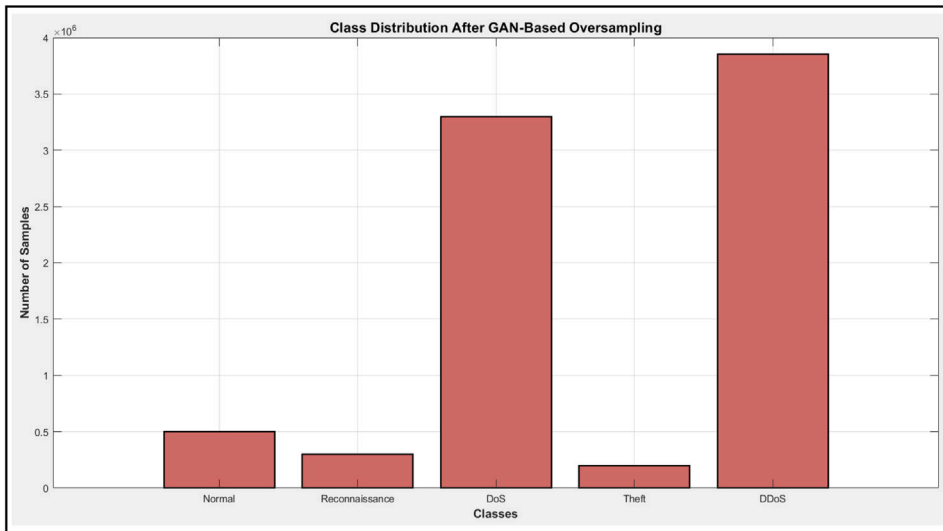


Fig. 6. Class-wise sample distribution after GAN-based oversampling for Bot-IoT dataset.

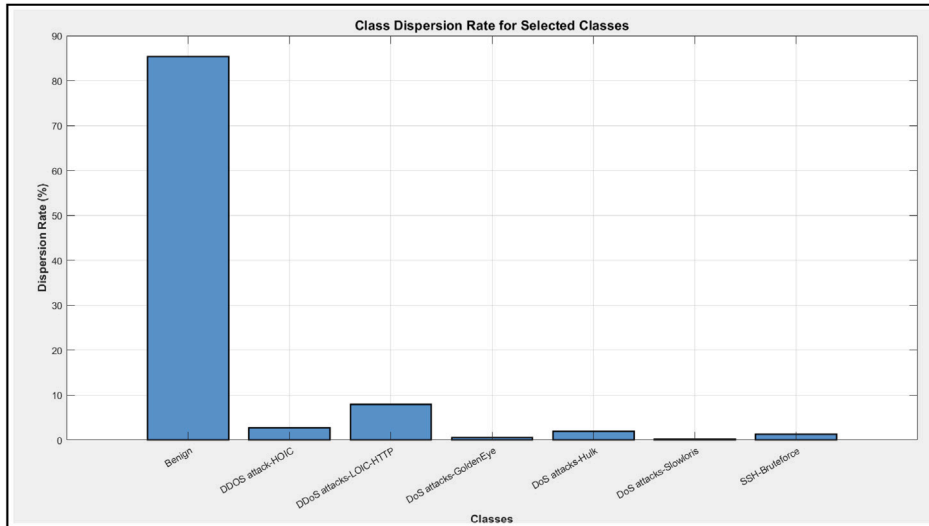


Fig. 7. Class-wise sample distribution from CSE-CIC-IDS2018 Dataset.

test was used to compare the actual class frequencies in the 10% sample with the expected frequencies based on the proportions of the full dataset, to assess whether this smaller dataset accurately reflects the class distribution of the entire dataset. The statistical analysis yielded a Chi-square statistic of 5.4513 and a p -value of 0.2440, which is well above the 0.05 significance level. The Chi-square test indicated no significant distributional differences, confirming the representativeness of the sample. Therefore, it is concluded that the 10% sample is a valid and representative subset for further research, as it maintains the overall class proportions of the original GAN-generated dataset.

The interoperability of DL framework is ensured, and the attributes are transformed into vector notation. This DENNW network is composed of several layers that have been developed to deal with the intricacies of IoT-cloud data. The Adam optimizer is used, with a batch size of 150 and a learning rate of 0.001. Due to computational limitations, training is restricted to 10 epochs. Table 8 discusses the detailed hyperparameter exploration of different techniques (NCA, GAN, DENNW) employed in executing this framework. A confusion matrix has been utilized to analyze the training model, giving insight into its class-wise effectiveness. The capacity of the model to accommodate both dominant and under-represented classes has been proved by estimating metrics including accuracy, precision, recall, and F1-score for every class. The suggested strategy presents a structured framework for tackling class imbalance and boosting the quality of detection in IoT-cloud intrusion situations. The performance of the framework is demonstrated in the subsequent section by means of the explanation and evaluation of the experimental results.

Table 9

Performance metrics against different attack types for the IoT domain.

Author	Metric	Normal	Reconn- aissance	DoS	Theft	DDoS	No. of features	Algorithms used	Classified classes	Overall
Present Study	Precision	0.6115	0.9982	0.9878	0.0000	0.9648	15	DENNW	5	–
	Recall	0.9978	0.4925	0.9571	0.0000	1.0000				–
	F1-Score	0.7583	0.6596	0.9722	0.0000	0.9821				–
	Accuracy	0.9978	0.4925	0.9571	0.0000	1.0000				0.93
[110]	Precision	–	–	–	–	–	10	SVM-LSTM	1	0.99
	Recall	–	–	–	–	–				0.99
	Accuracy	–	–	–	–	–				0.99
[116]	Accuracy	–	–	–	–	–	26/27	Proposed (IIDS-IoT)	Binary- 3,5,11	99.60,98.07- 98.33,98.31
[111]	Accuracy	–	–	–	–	–	26/27	RF/SVM	Binary-Multi	98.33%, 99.60%/ 98.31%, 98.07%
[114]	AUC	–	–	–	0.98785%	–	Full	Light-GBM	Binary	98.75%
[113]	Accuracy	–	–	–	–	–	Full	PB-DID, LSTM	3	96.3%
[112]	Precision	–	–	–	–	–	16	CNN	Binary	0.9827
	Recall	–	–	–	–	–				0.9819
	Accuracy	–	–	–	–	–				0.9819
[115]	Recall	1.00	1.00	0.99	1.00	0.99	Full	ONIDS	14	–

5. Results and discussion

The present study demonstrates strong performance in multi-class IoT intrusion detection, specifically in classifying five distinct attack types: Normal, Reconnaissance, DoS, Theft, and DDoS. This robust implementation is performed by employing a DENNW with only 15 features. Hence, it is more computationally effective compared to other techniques discussed in Table 9 like IIDS-IoT, Random Forest (RF)/Support Vector Machine (SVM), and ONIDS, which necessitate full feature sets for satisfactory performance. Although binary classifiers such as [110,111], and [112] attain higher accuracy rates (approximately 98%–99%). They are not suitable for real-life multi-class threat evaluation considering their restricted binary classification scope. The present research retains an extraordinary 93% overall accuracy regardless of managing multiple attack categories, with robust efficiency in DoS (97.22% F1-score) and DDoS (98.21% F1-score). At the same time, it strives for Theft attack detection (0% F1-score and recall). This is due to the highly imbalanced nature of the data. However, it dominates models like [113] (96.3% accuracy for only three classes) and [114] (LightGBM with 98.785% AUC for binary classification) by presenting a more stable and scalable methodology.

Conversely, [115] (ONIDS) includes 14 classes but demonstrates recall values of 1.00 for some classes. This may indicate a class imbalance or probable overfitting, resulting in a lack of generalization in real-world settings. The current investigation presents better precision in class-wise F1 scores and recall metrics, rendering it a more trustworthy choice for a practical environment. Furthermore, models such as [116] (IIDS-IoT) and [111] (RF/SVM) achieve over 98% accuracy, they depend on a larger feature set (26/27), which improves the computational cost. Figs. 8 and 9 depicts that the current work achieves a balance between competence and multi-class generalization, making it a feasible alternative for real-time intrusion detection in the IoT domain. Furthermore, Theft attack detection exhibits that additional advancements are required. The main cause of the 0% detection rate for theft assaults is the extremely small number of actual instances (139) used for GAN training. This limited the development of comprehensive features and caused the method to fail, resulting in artificial specimens showing low intra-class variability. Additionally, it becomes more difficult to distinguish between the two categories, Normal and Reconnaissance, due to the significant similarities in the statistical distribution of Theft traffic. As a result, the algorithm may have focused more on GAN-specific features rather than genuine Theft signals, leading to a considerable decrease in effectiveness for theft category. Conclusively, this methodology remains competitive in terms of accuracy, class-wise detection, and resource efficiency compared to existing studies.

5.1. Cloud domain

Figs. 10 and 11 demonstrate the effectiveness of the present study regarding the use of a Deep Ensemble Neural Network (DENNW) for detecting multiple types of attacks. Table 10 presents that this DENNW attained an overall accuracy of 97.22% across seven classified classes. The framework outperforms in identifying benign traffic, DDoS attacks (HOIC and LOIC-HTTP), and SSH-Bruteforce, with extraordinary precision, recall, and F1-scores close to 1.00%. However, the detection rate somewhat drops for DoS-based attacks like GoldenEye (recall: 0.3128, F1-score: 0.4684) and Hulk (recall: 0.1425, F1-score: 0.2411). This specifies that although the model behaves efficiently in classifying some attack types, further optimization may be needed for fine-grained classification of DoS variants. Overall, the model exploits full feature selection to guarantee widespread attack detection and surpasses several prevailing methodologies concerning multi-class classification capacity. In contrast to previous studies, the current research obtains a higher overall accuracy (97.22%) than [117] (91.40% to 95.45%) and [118] (92.07% to 99.77%), exposing overwhelming behavior across different attack types.

However, [119] (Stacked Autoencoder (SAE)+ Apache Spark (ASpark)) attains absolute precision and recall for the majority of attack types. Its categorization is regulated and overall accuracy is lower at 93%. [120] (Gradient Boosting) achieves 99% accuracy, marginally surpassing the present study in overall classification; however, it does not offer detailed class-wise metrics, making it

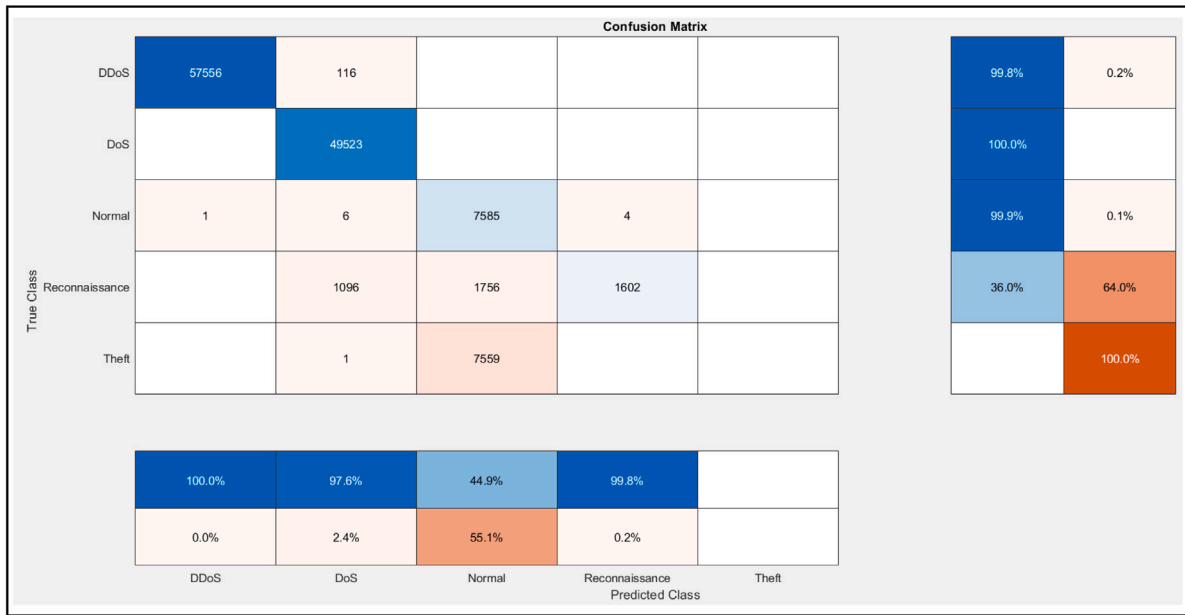


Fig. 8. Confusion Matrix generated by performing training against the proposed GAN-based Deep Ensemble Neural Network for the Bot-IoT dataset.

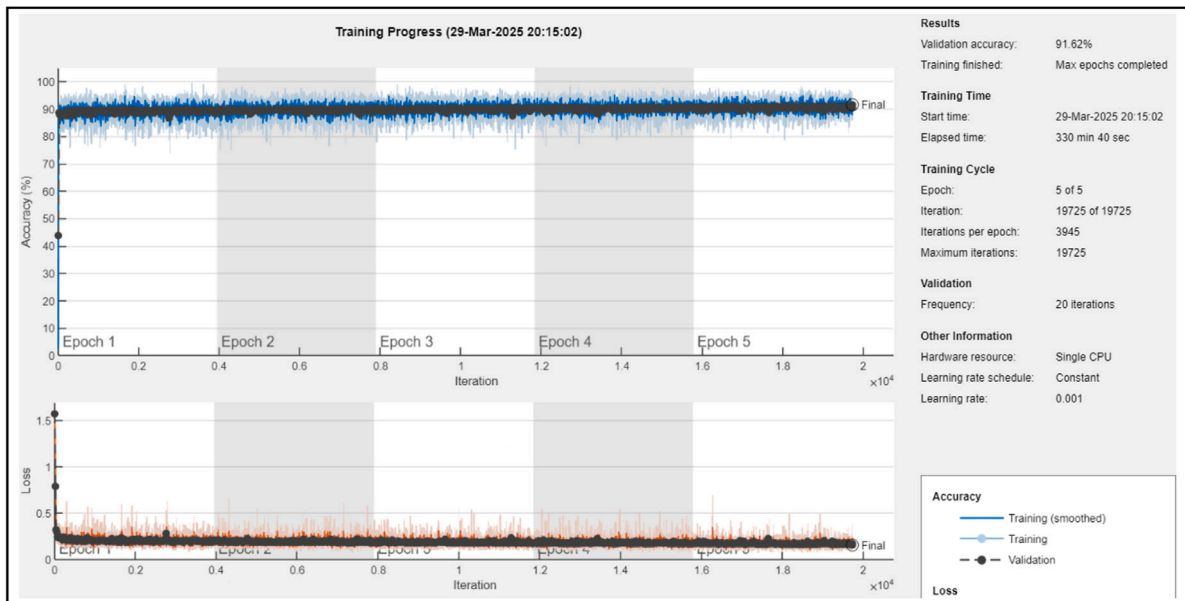


Fig. 9. Confusion Matrix generated by performing training against proposed GAN-based Deep Ensemble Neural Network for Bot-IoT dataset.

challenging to evaluate distinct attack performance. Significantly, the present study outperforms prevailing techniques in terms of DL amalgamation, feature exploitation, and attack coverage, devising it a strong candidate for real-world cybersecurity applications.

The IoT and Cloud datasets have separate threat categories and label classifications, making straightforward migration analysis difficult. However, this research effectively examined the adaptability of DENNW to domain shifts, which is the main focus of this study. The ability of DENNW to handle variations in traffic behavior, features, and disturbances was tested by using each dataset independently, applying the same preprocessing, feature normalization, and denoising-weighting techniques. The consistent and stable performance across these two very different contexts shows that DENNW can generalize and adapt effectively, regardless of differing class structures.

Table 10
Class-wise performance analysis for cloud domain.

Author	Metric	Benign	DDoS-HOIC	DDoS-LOIC-HTTP	DoS-GoldenEye	DoS-Hulk	DoS-Slowloris	SSH-Brute-force	No. of features	Algorithms used	Classified Classes	Overall
Present study	Precision	0.9761	0.9878	0.9436	0.9319	0.7821	0.8189	0.9794	Full	DENNW	7	–
	Recall	0.9952	1.0000	0.9727	0.3128	0.1425	0.7027	0.9972				–
	F1-Score	0.9855	0.9939	0.9580	0.4684	0.2411	0.7564	0.9882				–
	Accuracy	0.9952	1.0000	0.9727	0.3128	0.1425	0.7027	0.9972				0.9722
[117]	Precision	–	–	–	–	–	–	–	Full	ANN	Multi	0.9545
	Recall	–	–	–	–	–	–	–				0.9140
	Accuracy	–	–	–	–	–	–	–				0.9543
[118]	Precision	–	–	–	–	–	–	–	11	XGBoost	Multi	0.920757
	Recall	–	–	–	–	–	–	–				0.98879
	F1-Score	–	–	–	–	–	–	–				0.949388
	Accuracy	–	–	–	–	–	–	–				0.997706
[119]	Precision	0.95	1.00	1.00	1.00	1.00	1.00	0.67	Full	SAE	7	–
	Recall	0.99	1.00	1.00	1.00	1.00	1.00	0.93				–
	F1-Score	0.97	1.00	1.00	1.00	1.00	1.00	0.78				–
	Accuracy	–	–	–	–	–	–	–				0.93
[120]	Precision	–	–	–	–	–	–	–	Full	GBoost	7	0.99
	Recall	–	–	–	–	–	–	–				0.98
	Recall	–	–	–	–	–	–	–				0.98
	F1-score	–	–	–	–	–	–	–				0.99
	Accuracy	–	–	–	–	–	–	–				0.98

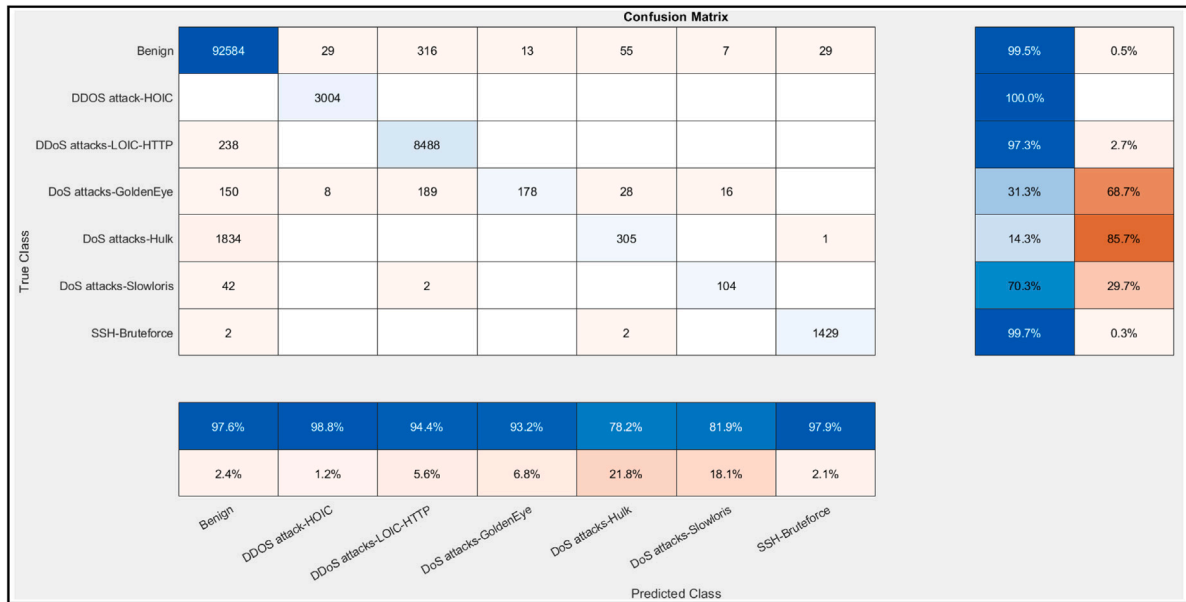


Fig. 10. Confusion Matrix generated by performing training against Deep Ensemble Neural Network for CSE-CIC-IDS2018 dataset.

6. Limitations and challenges

Exploring ML and DL models, cyberattacks, and datasets for IoT and cloud environments inherent limitations and challenges. The rapidly progressing feature of IoT and cloud-based technology demands consistent modifications and adaptations. Moreover, following the diversity and intricacy of IoT devices and cloud platforms, standardization and compatibility issues lead to instability in the datasets used for model training. The deprivation of fully ground truth-labeled datasets for cybersecurity-related tasks that are comprehensible to the public is another considerable challenge. Data privacy, ownership, and consent-related ethical issues obfuscate data collection, distribution, and use in cloud and IoT ecosystems. Finally, it emphasizes the requirement for coordination by requiring expertise in other domains.

7. Conclusion

The proposed technique facilitates systematic development of intrusion detection systems designed for IoT–cloud infrastructures while offering beneficial knowledge for network administrators and system architects. The methodology can be deployed in real-life situations to enhance adaptability against emerging cyberthreats by employing deep learning and class-aware augmentation. This

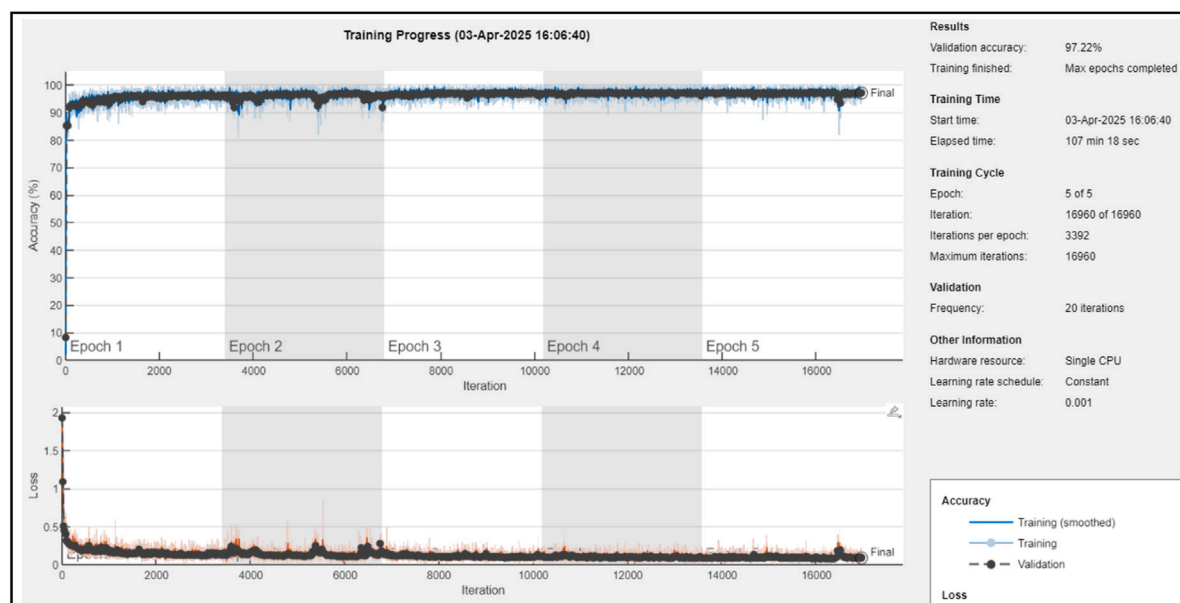


Fig. 11. Confusion Matrix generated by performing training against Deep Ensemble Neural Network for CSE-CIC-IDS2018 dataset.

research thoroughly analyzes the existing literature to showcase the multifarious nature of cyberattacks that these interconnected domains face. The current study has detected significant shortcomings in the prevailing literature, particularly concerning the development of such DL-based frameworks that can efficiently manage the distinctive security concerns offered by the amalgamated IoT-cloud domain. Hence, an intelligent intrusion detection system has been demonstrated in this research to deal with class imbalance in IoT-cloud frameworks by utilizing DL models and GAN-based data integration. This proposed framework has been trained and analyzed using cross-domain datasets, namely Bot-IoT and CSE-CIC-IDS2018. The model achieved excellent detection rates with optimized class-wise metric analysis by employing GANs to combine minority class instances and targeted class fusion. The recommended ensemble deep neural network prevailed over many existing techniques in terms of accuracy, precision, recall, and F1-score. Furthermore, it surpassed preceding studies that emphasized binary classification or missed significant metric breakdowns, achieving 97.22% overall accuracy in cloud-based assessments and 93% in IoT situations with full class-wise metric evaluation.

The goal for future research is to assess the proposed structure for more robust GAN variants. These structures could potentially address the challenges identified in this work, especially the 0% detection rate for the Theft class, and would demonstrate improved resilience and sample diversity across previous studies. This type of investigation will enhance the versatility and reliability of GAN-augmented intrusion detection in various cyberattack scenarios. Furthermore, to improve privacy protection over scattered IoT-cloud networks, the framework can be strengthened for future research by including federated learning for dispersed scenarios. Moreover, the framework would be a better option in environments with sparse assets if it existed in real-time and were deployed inexpensively on peripheral endpoints. Implementing explainable AI (XAI) approaches could potentially improve interpretability and reliability of the decision-making process, specifically in key sectors like medical and smart transit. Additionally, the resiliency of the recommended strategy can be evaluated and extended by widening its analysis across several baseline datasets and actual traffic.

CRedit authorship contribution statement

Sadia Nazim: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Syed Shujaa Hussain:** Writing – review & editing, Visualization, Validation, Software, Resources, Formal analysis. **Bilal Yousuf:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation. **Saima Sultana:** Writing – review & editing, Validation, Resources, Project administration, Investigation, Formal analysis, Data curation. **Eraj Tanweer:** Writing – review & editing, Visualization, Validation, Software, Resources, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed in this study are available at the provided links.

<https://research.unsw.edu.au/projects/bot-iot-dataset>

<https://www.unb.ca/cic/datasets/ids-2018.html>.

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